TARESH BANSAL

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RESEARCH INTERESTS

AI for Science · Trustworthy Medical AI · Multimodal AI · AI Agents · Vision-Language Models

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani, India Aug. 2025 – May 2027
Master of Engineering in Computer Science **GPA: 8.68 / 10.0**

Birla Institute of Technology and Science (BITS), Pilani, India Nov. 2020 – May 2025
Bachelor of Engineering in Computer Science & M.Sc. in Chemistry (dual degree) **GPA: 8.13 / 10.0**

PUBLICATIONS

A Novel Competing Convolutional Kernels Method to CSI-Based Fall Detection for Disabled People [DOI]
Siva Sai, **Taresh Bansal**, Vinay Chamola
IEEE Internet of Things Magazine Nov. 2025

Drift-Resilient Intrusion Detection in IoMT Networks Using Adaptive Hoeffding Trees with Internal Node Reassessment
Taresh Bansal, Siva Sai, Vinay Chamola, Mohan Gurusamy
IEEE Internet of Things Magazine, under review Sep. 2026

RESEARCH EXPERIENCE

Do Medical VLMs Read the Image? Auditing Chest X-ray Reports Sep. 2026 – Present

- Asked whether AI models writing chest X-ray reports actually read the image, which grading can't tell.
- Swapped in another patient's X-ray, a black image or no image across **5** models and **3** datasets.
- Confirmed, as preregistered, that X-ray specialists use the image more than medical generalists.
- Showed grading misses this: Lingshu scores **1.29** with or without the image, vs MAIRA-2's **1.36**.
- Found that no model flags a black image as unreadable; each still writes a normal-sounding report.

Visual Token Pruning for Medical Vision-Language Models Sep. 2026 – Present

- Reproduced CDPruner (NeurIPS 2025) on two open models, matching all **24** scores within **0.32 points**.
- On LLaVA-Med, it beat random selection by **3.9 points** on SLAKE closed questions ($p = 0.004$).

Ask My Warehouse — LLM Agent Aug. – Sep. 2026

- Cleaned raw retail sales data into a star-schema warehouse with **PySpark**, then analysed it in **SQL**.
- Built an **MCP** server from scratch that gives an LLM safe, read-only access to the warehouse database.
- Built a **LangChain** agent on top of it that turns plain-English questions into SQL and explains results.

Lightweight Architectures for Plant Disease Recognition on Edge Devices Mar. – Aug. 2026

- Compared **9** small models for on-phone crop disease detection across **3** datasets and **118k** images.
- Found the proposed model (**2.03M** parameters) best overall, with top accuracy on **2 of 3** datasets.
- Led the next model by **3.8 points** on the hardest, 115-class dataset (**69.8%** against **66.0%**).
- Showed CBAM attention hurt rather than helped, costing **3.4 points** of accuracy on the hardest dataset.
- Measured **10.5 ms** per image over **200** runs, **40%** faster than the other Ghost-based models (17–18 ms).

Drift-Resilient Intrusion Detection for IoMT Networks Jan. – Feb. 2026

- Tackled attack detection on medical IoT devices, where traffic shifts over time and models go stale.
- Trained on the WUSTL-EHMS-2020 dataset, balancing its attack classes with **SMOTE** after cleaning.
- Reached **99.94%** accuracy, **9.05 points** above the best incremental baseline, at **100%** precision.
- Updated in **0.65 s** per cycle against **41.5 s** to retrain NuSVC and **199 s** for Isolation Forest.

Flood Detection from Drone Imagery Nov. – Dec. 2025

- Trained ResNet, InceptionNet and Xception to spot flooding in drone photos, reusing pretrained layers.
- Tuned the training settings (BCEWithLogitsLoss, Adam) to reach **99.1%** accuracy.

Image Inpainting with GANs and Autoencoders Feb. – Apr. 2025

- Filled missing image regions with MI-GAN and a custom autoencoder: **92.3%** and **84.9%** accuracy.

TEACHING EXPERIENCE

Teaching Assistant, BITS Pilani Aug. 2024 – Present

Introduction to Logic, BITS Pilani Digital	Sep. 2026 – Present
Introduction to LLMs, BITS Pilani	Aug. 2026 – Present
Computing Systems, BITS Pilani Digital	May 2026 – Aug. 2026
Introduction to Computing Systems, BITS Pilani Digital	Dec. 2025 – Mar. 2026
Object-Oriented Programming, BITS Pilani Digital	Sep. 2025 – Dec. 2025
Foundations of Data Science, BITS Pilani	Aug. 2025 – Dec. 2025
Machine Learning, BITS Pilani	Aug. 2024 – Dec. 2024

POSITIONS OF RESPONSIBILITY

Executive Committee Member, National Service Scheme (NSS) Aug. 2021 – July 2022
BITS Pilani, Pilani, India

- Managed village health and eye camps with a team of 20+ volunteers, serving 250+ attendees.
- Led a mental-health and technology-awareness lecture series for 180+ students with 30+ volunteers.
- Started the "Health OK Please" podcast on cervical cancer and emergency care, reaching 100+ people.

Event Coordinator, Chemistry Association Aug. 2021 – Apr. 2022
BITS Pilani, Pilani, India

- Ran **four** live events at APOGEE, the annual tech-fest: **1500+** registrations, **200%** more footfall.
- Tripled inter-college reach and collaborations by spacing events evenly, leading **40+** volunteers.

Anti-COVID Force, BITS Pilani Apr. 2021 – Aug. 2021

- Coordinated 100+ volunteers who resolved 400+ COVID-19 queries on oxygen, beds and medical facilities.

PROFESSIONAL EXPERIENCE

NLP Intern, Palmtree Infotech May 2022 – July 2022

- Built a language-conversion tool from English to Hebrew and other languages using Python and spaCy.
- Implemented NLP techniques to extract the key content of documents for use in multilingual settings.

TECHNICAL SKILLS

Languages	Python, C++, SQL
Machine Learning	PyTorch, Computer Vision
Data Processing	NumPy, Pandas, PySpark
LLM Tooling	LangChain, MCP, Pydantic

RELEVANT COURSEWORK

Machine Learning, Deep Learning, Data Mining, Probability and Statistics, Data Structures and Algorithms, Distributed Systems, Operating Systems, Database Systems, Object-Oriented Programming.

HONORS AND ACHIEVEMENTS

2025: Qualified the Graduate Aptitude Test in Engineering (GATE) in Data Science and AI.